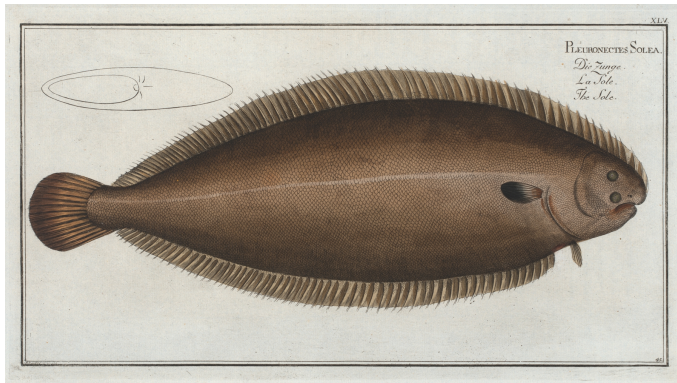




Reconstruction of past total discards for North sea sole (sol.27.4)

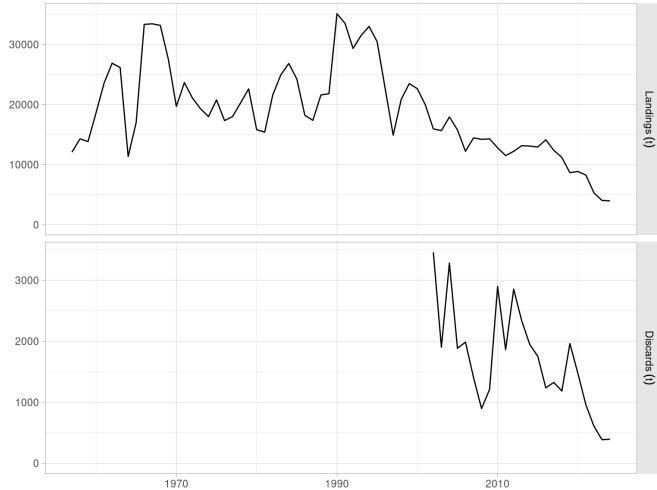
ICES WGNSSK 2026

21-30 April 2026 | Iago Mosqueira (Wageningen Marine Research)

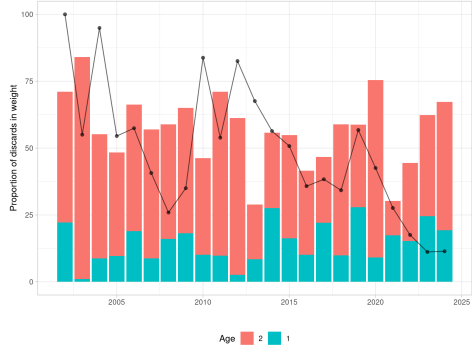
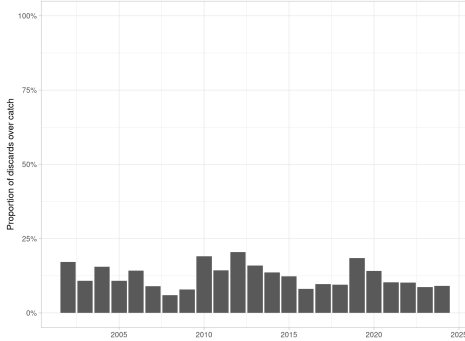
Discards reconstruction for sol.27.4

- Previous assessment (AAP, Aarts & Poos, 2009) estimated discards prior to 2002
- Discards part of the system equation
- Separated landings & discards fleet in SS3 to apply different wtatage
- Makes discards reconstruction not possible in SS3
- Reconstruction outside of assessment model to evaluate impact
- *Reconstruct discards back in time for the period prior to 2002, where estimates are lacking.* WGNSSK, 2024.

Data - Available landings and discards



Data - Discards proportions



Methods - discard ratio

- Compute average discards to landings ratio, 2002-2006
- Apply to landings 1959-2001 to compute discards

Methods - GLM

$$\text{Discards}_i \sim \text{Gamma}(\mu_i, \theta)$$

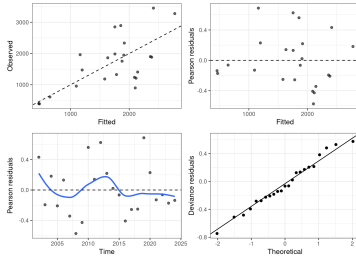
$$\log(\mu_i) = \beta_0 + \beta_1 \log(\text{Landings}_i) + \beta_2 \log(\text{Rec}_{i-1}) + \beta_3 \log(\text{Rec}_{i-2})$$

```
mod12 <- glm(discards ~ log(landings) + log(rec1) + log(rec2),  
             family = Gamma(link = "log"))
```

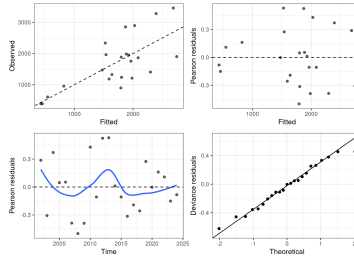
- `discards ~ landings`
- `discards ~ landings + R12`
- `discards ~ landings + R`

Diagnostics

Discards ~ Landings

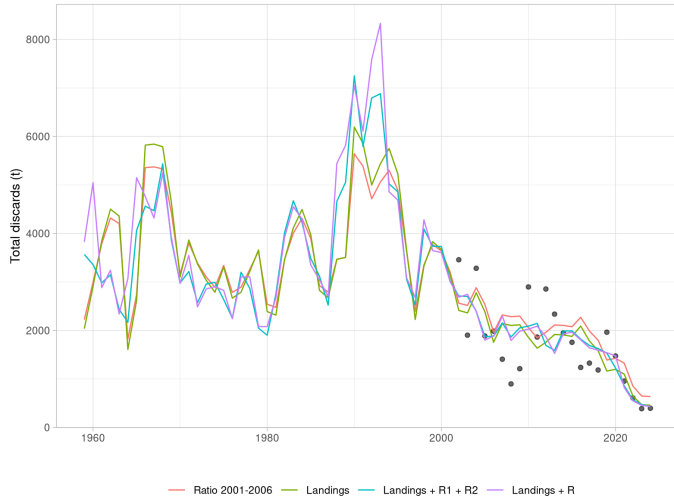


Discards ~ Landings + R12

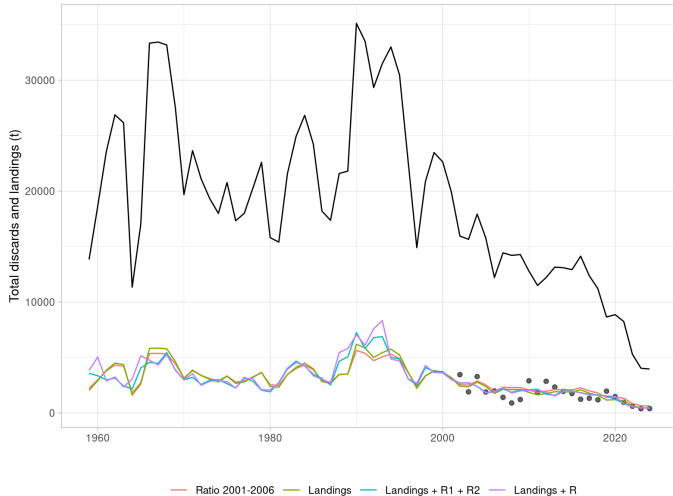


Model	DF	AIC	BIC
Landings	3	357.9	361.3
Landings + Rec1 + Rec2	5	357.2	362.9
Landings + Rec	4	353.8	358.4

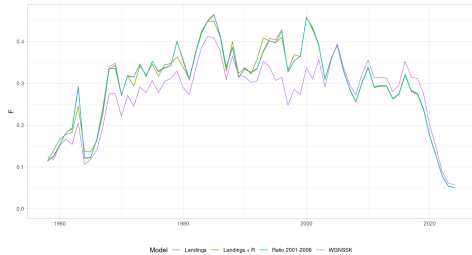
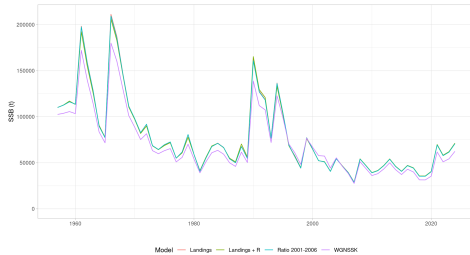
Results - Discards predictions



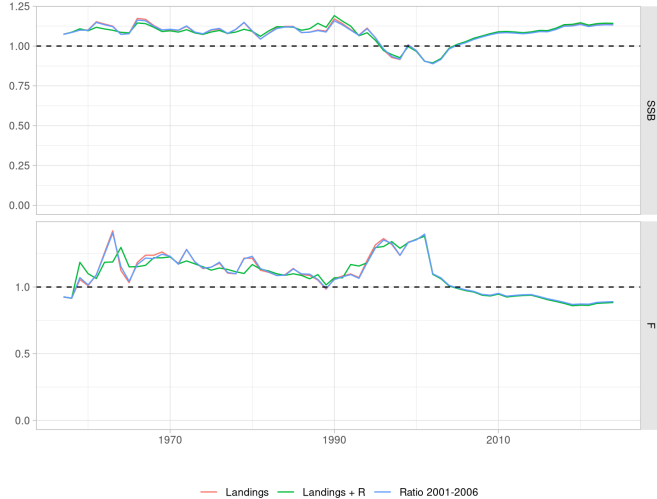
Results - Discards predictions + landings



Results - SSB and F



Results - SSB and F compared to WGNSSK

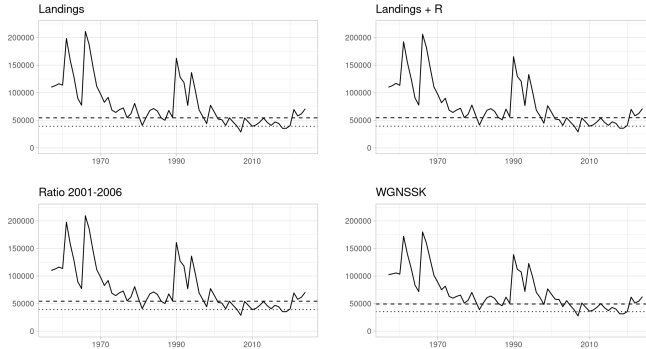


Results - reference points

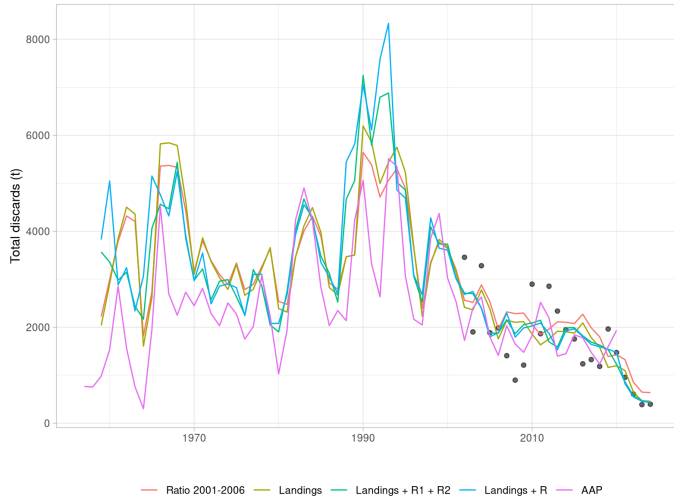
Reference point	WGNSSK*	Landings	Landings + R	Ratio 2001-2006
$MSYB_{trigger}$	49265	54476	54696	54316
F_{MSY}	0.19	0.217	0.214	0.215
B_{lim}	35453	39203	39362	39089

*WGNSSK refers to the recomputed values using the 1957:2024 dataset.

Results - SSB trajectories and reference points



Results - discards predictions including AAP (2021)



https://github.com/iagomosqueira/2026_sol.27.4_review/discards_reconstruct

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